When can tuned simulated annealing be beaten at tensor-network contraction ordering?

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Abstract

Simulated-annealing tree refiners are the strongest general-purpose optimizers of tensor-network contraction orders, but published comparisons rarely establish when a tuned annealer can be beaten at all. We answer this question on ten benchmark networks—random regular graphs, quantum circuits including the real 53-qubit Sycamore network, surface-code syndrome networks, and counting and inference networks—under a hardened protocol: matched single-threaded budgets, independent rescoring of every tree from its topology, per-run instance relabeling, and confirmation-based records. Four results emerge. First, tuning dominates mechanism design: one shipped default (a memory target of 2^{20}) costs 20× to 2100× in flops, and until it is fixed, no comparison between optimizers is meaningful. Second, on small converged instances the tuned annealer cannot be beaten: thirteen mechanically distinct search paradigms and an independent toolchain tie its frontier, which we certify width-optimal with the optimum localized to $[53, 61.5]$ on the circuit proxy, and we prove that no bound computable from the isoperimetric profile can close the interval. Third, on large structured instances the annealer is beaten by two mechanisms that inject global information as *input* rather than as search control: rank-non-increasing simplification (which removes up to 89% of tensors and yields a 2^{16} improvement on Sycamore) and waist surgery (bounded Fiduccia–Mattheyses improvement of the dominant contraction's bipartition, which sets records on three instances). Behind the latter lies a structural finding: across hundreds of instrumented calls the annealer's dominant cut is *never* a locally minimal balanced cut. Fourth, on probabilistic-inference networks the frontier splits into two regimes—elimination methods win dense Bayesian networks while annealing wins pedigree linkage—and shipped defaults of an inference package sit 12–30 bits above that frontier. All records survive a same-machine matched-budget re-measurement against the reference Julia implementation, and every claim ships with machine-checkable artifacts from the 56-attempt steered autonomous research loop that produced it.

1 Introduction

The cost of contracting a tensor network is governed by the order in which its tensors are combined. Finding a good order is the computational bottleneck of classical random-circuit simulation [8, 7], of quantum error-correction decoding, and of exact probabilistic inference, and a rich ecosystem of optimizers has grown around the problem: simulated-annealing tree refiners [5], hyper-optimized portfolios over greedy and hypergraph-partitioning constructions [3, 12], treewidth machinery from the parameterized-algorithms community [2, 14], and exact methods for small networks [10]. Among these, tuned simulated annealing over contraction trees is in practice the strongest general-purpose refiner. Yet the field's comparisons are relative—an optimizer is good if it beats another optimizer—and they leave the practical question unanswered: *when, and by what kind of mechanism, can a well-tuned annealer actually be beaten?*

This paper answers that question empirically and, where possible, certifiably. The answer has four parts, in increasing order of what they demand from a challenger. (i) Before any mechanism

instance	tensors	indices	provenance
reg3_250	250	372	synthetic, closed (certification study)
sycamore_m20	561	963	synthetic Sycamore-like proxy (certification)
sycamore_53_20_0	3369	—	real Sycamore 53q, m=20 [3]
surfacecode_d9–d21	219–2203	—	surface-code syndrome networks
ksg	5197	—	king-graph independent-set counting
reg3_1000	1000	1500	random 3-regular, n=1000
rqc_97_m24	1238	—	97-qubit random circuit
nqueens_28	4252	—	28-queens counting
dbn_13	572	44 labels	deep-belief-network inference (UAI DBN_13)
qft_27	405	—	27-qubit QFT, 27 open indices

Table 1: Benchmark instances. The first two are the closed synthetic networks of the certification study (Sec. 4); the rest are imported real-provenance benchmark networks. Ten additional UAI-2014 inference instances are introduced in Sec. 8.

matters, the incumbent must be tuned: a single shipped default costs more than every algorithmic effect we measured combined (Sec. 3). (ii) On small instances where the tuned annealer converges, it cannot be beaten at all—we certify its frontier width-optimal and prove an impossibility result for an entire family of lower-bound techniques (Sec. 4). (iii) On large structured instances it *can* be beaten, by exactly two mechanism families, both of which inject global structure as input to the local search rather than replacing its control loop: structural simplification (Sec. 5) and waist surgery (Sec. 6). (iv) On probabilistic-inference networks the answer bifurcates by graph structure: annealing itself is beaten by elimination orderings on dense Bayesian networks, while annealing wins pedigree-linkage networks by equally large margins (Sec. 8).

Every comparison behind these claims is matched-budget and adversarially validated (Sec. 2), the records are re-measured on a single quiet machine against the reference Julia implementation (Sec. 7), and the negative results—twenty falsified mechanisms with measured root causes—are reported with the same care as the wins (Sec. 10). Figures 1 and 8 summarize the tuning result and the record board; the reader who consults only those two figures and Table 2 will have the paper's core content.

2 Setup and evaluation protocol

Instances and cost model. A tensor network is a hypergraph $G = (V, E)$: vertices are tensors and (hyper)edges are indices; contracting to a scalar (or a small open-index set) requires a full binary contraction tree over the tensors. Writing cost_v for the $\log_2$ floating-point cost of internal node v (the sum of $\log_2$ dimensions over the union of the operands' indices), the two standard quality measures are the total time cost

$$\text{tc} \;=\; \log_2 \sum_v 2^{\text{cost}_v}, \tag{1}$$

a log-sum-exp over all $|V| - 1$ contractions, and the space cost sc, the $\log_2$ size of the largest intermediate tensor. All numbers in this paper are $\log_2$ quantities; a difference of 1.0 is a factor of two in flops.

Table 1 lists the ten benchmark instances. Two are closed synthetic networks used for the certification study (a 250-tensor random 3-regular graph and a 561-tensor Sycamore-*like* circuit proxy, which we distinguish explicitly from the real circuit). The remainder are imported from the OMEinsumContractionOrders benchmark suite [6] with provenance recorded at import: the real Sycamore 53-qubit, 20-cycle network, surface-code syndrome networks at distance $d = 9 \ldots 21$, an independent-set counting network on a king graph, a 1000-vertex random 3-regular graph, a 97-qubit random-circuit network, the $n = 28$ queens counting problem, a deep-belief-network inference instance, and a 27-qubit quantum Fourier transform. Section 8 adds ten hard instances from the UAI-2014 inference competition, exported directly from the artifact shipped with TensorInference.jl.

Protocol. Every optimizer run gets 90 seconds of single-threaded wall clock per instance unless stated otherwise. Emitted trees are rescored by an independent checker that recomputes tc and sc from the tree topology and the original index sets alone; optimizer-reported numbers are never trusted. Instances are randomly relabeled per run, so memorized answers keyed on input encoding do not transfer. A best-known value ("record") only moves when a run beats the incumbent by more than 0.05 and a second run under fresh relabeling confirms; the worse of the two runs is recorded. On instances above 1000 tensors, where single-run variance exceeds method differences, scored values are medians of three runs. Per-child CPU-time accounting rejects hidden parallelism. Reference rows (the tuned annealer of Sec. 3, and cotengra's hyper-optimizer and SA refiner [3]) are measured best-of-three with the identical pipeline.

Same-machine re-measurement. Because the record board accumulated on a development machine, all headline comparisons were re-measured from scratch on a single dedicated 2-core Linux server: 305 scored runs covering budget scaling (90/300/900 s, three repetitions), run distributions (15 repetitions), and the surface-code family (five repetitions), plus a matched-budget ladder of the reference Julia implementation (OMEinsumContractionOrders.jl: TreeSA at both memory-target settings and one/four/eight restarts, HyperND, and three treewidth back-ends) run through that benchmark suite's own harness, and the inference batch of Sec. 8. Every cross-method statement in Secs. 7 and 8 is same-machine and matched-budget.

3 Tuning before beating: a mis-set default dominates all algorithmic effects

The TreeSA-family annealers [5] optimize a composite objective in which the space cost is pulled toward a target $\mathrm{sc}_{\mathrm{target}}$, with a penalty active whenever $\mathrm{sc} > \mathrm{sc}_{\mathrm{target}}$. The implementation we study (omeco, a Rust re-implementation aligned with OMEinsumContractionOrders.jl) inherited the shipped default $\mathrm{sc}_{\mathrm{target}} = 20$. On instances whose good trees have naturally larger intermediates—the optimum-quality trees on our two closed instances have $\mathrm{sc} = 34$ and 53—this default is not merely unhelpful. It actively drags the search into a low-memory, high-flop region of tree space.

Figure 1 shows the resulting tc at a fixed 90-second budget as a function of $\mathrm{sc}_{\mathrm{target}}$. **This figure is our first main result**, and the transition it reveals is a cliff, not a slope: every target below the natural space cost of good trees is roughly equally harmful (medians 45.6–47.0 on reg3_250, 71–77 on sycamore_m20), quality snaps to the frontier exactly at the natural value ($\mathrm{sc}_{\mathrm{target}} = 34$ and 53), and stays there unchanged through $\mathrm{sc}_{\mathrm{target}} = \infty$, i.e., with the memory objective deleted. The harm is strictly one-directional. At the shipped default the damage is 4.3 $\log_2$ units on reg3_250 and 9.7 on sycamore_m20 (20× and 800×); on the real Sycamore network it reaches 11.1 units (2100×, Sec. 7). The cliff is visible in *published* data as well: the benchmark suite of OMEinsumContractionOrders.jl reports Julia TreeSA at $\mathrm{tc} = 71.0$ on the real Sycamore network with its shipped $\mathrm{sc}_{\mathrm{target}} = 20$, within 1.6 of our re-measurement of the same configuration, and 11 units above the same annealer with the target removed.

The practical rule is: *never anneal toward a memory target below the natural space cost of good trees; absent a physical memory budget, use the natural value or no target at all.* Every comparison in the remainder of this paper is against the annealer with this fix applied—the *tuned reference*. Beating an untuned incumbent is not evidence of algorithmic progress.

4 Where the tuned annealer cannot be beaten: a certified null

Before exhibiting mechanisms that beat the tuned reference, we establish where none can. On the two closed instances the tuned annealer converges within the budget, and the question "can it be beaten?" admits a certified negative answer in three layers: empirical convergence, width-optimality, and an impossibility result for profile-based bounds.

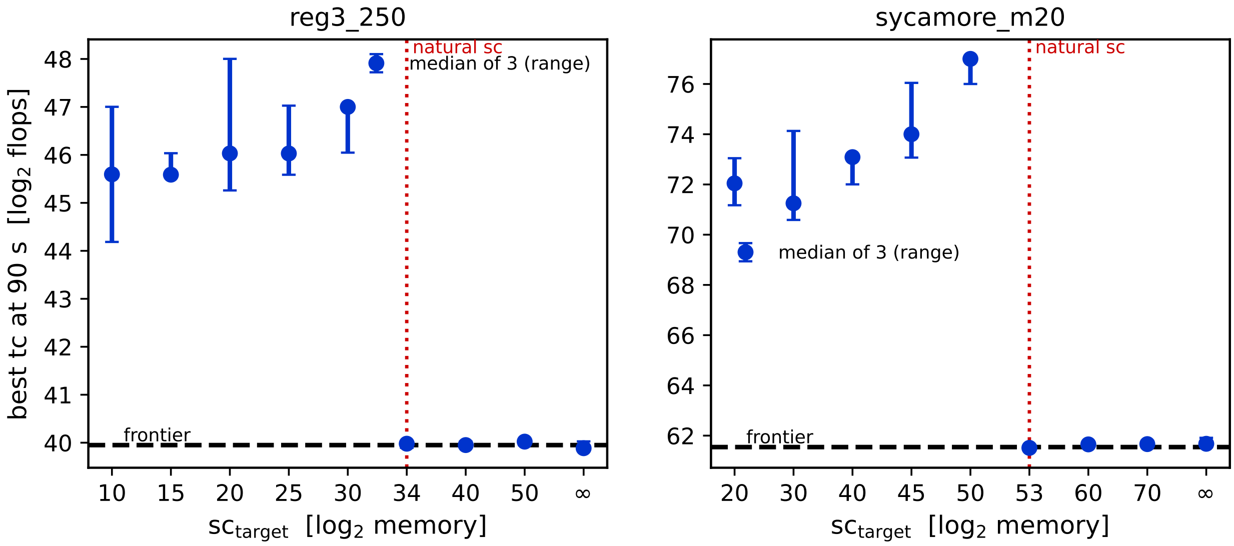

Figure 1: Best tc reached in 90 s of single-threaded annealing versus the memory target $\mathrm{sc}_{\mathrm{target}}$ (median of three runs; whiskers span the range). The dashed line marks the converged frontier of Sec. 4; the dotted vertical line marks the natural space cost of optimum-quality trees. The transition is a cliff at the natural value: every smaller target is comparably harmful, and every target at or above it, including $\mathrm{sc}_{\mathrm{target}} = \infty$, reaches the frontier. The shipped default is $\mathrm{sc}_{\mathrm{target}} = 20$.

4.1 Thirteen mechanisms, two toolchains, one frontier

We attempted to beat the tuned reference with mechanically distinct search paradigms, implemented independently and scored under the protocol of Sec. 2: coarse subtree-transfer moves and basin hopping [1], evolutionary replica populations, exact-penalty ramps, tabu search with strategic oscillation, large-neighborhood destroy-and-repair, beam search with lookahead, Boltzmann-greedy portfolios [3, 9], recursive balanced min-cut construction [13, 12], elimination-order annealing [2], order-then-tree interval dynamic programming [4], hierarchical coarsen-then-exact solves, structure-guided sweeps, and profile-aware annealing.

Figure 2 shows every scored result as its distance from the best verified value. All thirteen mechanisms land within 0.35 $\log_2$ of the tuned reference on both instances—most within noise—and none set a confirmed record. An independent toolchain reproduces the same frontier: cotengra's simulated-annealing refiner [3] ties it to within 0.05 and 0.14, while cotengra's hyper-optimization mode alone remains 1.7 and 2.4 behind at matched budget. Extending the budget tenfold moves the frontier by less than 0.1. At this scale the frontier is a property of the cost landscape, not of any search mechanism, schedule, or implementation.

4.2 Certified bounds: width-optimality and an impossibility result

Every classical lower-bound technique for contraction cost is *max-form*: it bounds the single largest contraction. The Markov–Shi correspondence [8] equates the minimal largest contraction rank with the treewidth of the line graph $L(G)$, and any balanced edge of a contraction tree yields a cut whose boundary lower-bounds the largest contraction.

On sycamore_m20 these tools give a sharp result. An explicit temporal cut severing all 53 qubit world-lines gives a balanced cut of value 53; extensive search finds nothing smaller, and the frontier trees achieve $\mathrm{sc} = 53$ exactly. The frontier is therefore *width-optimal*, and $\mathrm{tw}(L(G)) = 52$. Combined with the frontier value, **our second main result** is the certified localization

$$\mathrm{tc}_{\min}(\text{sycamore\_m20}) \in [53,\ 61.544]. \tag{2}$$

On reg3_250 the analogous certification is obstructed by the known hardness of certifying high

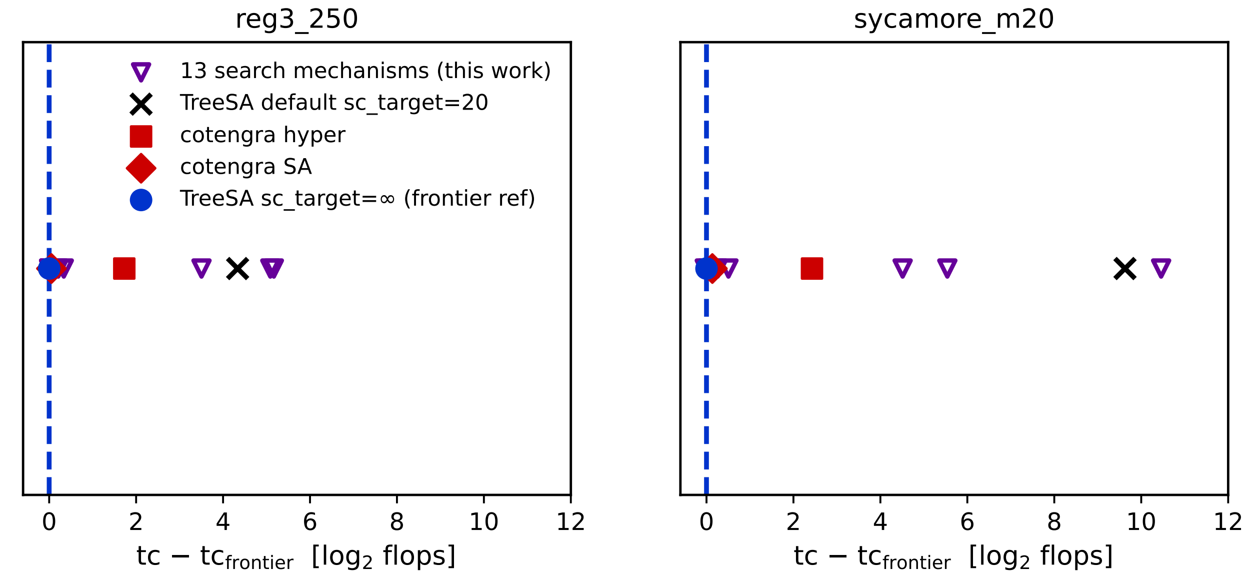

Figure 2: Distance of every scored optimizer from the frontier value (dashed line at zero). Open triangles: the thirteen search mechanisms of this work. Filled symbols: the tuned TreeSA reference ($\mathrm{sc}_{\mathrm{target}} = \infty$, circle), cotengra's SA refiner (diamond), cotengra's hyper-optimizer (square), and the shipped-default TreeSA (cross).

treewidth on expanders: the best machine-verified certificate (a 32-vertex minor of $L(G)$, checked by two independent exact solvers) gives $\mathrm{tw}(L(G)) \geq 18$ against a constructive upper bound of 34.

The width bound has a ceiling: tc is a log-sum (1) over $\sim |V|$ contractions, and the frontier's excess over the width bound—8.5 on sycamore_m20, close to the maximal possible $\log_2 561 = 9.1$—is invisible to any bound on a single contraction. Closing the interval (2) from below requires *sum-form* bounds. For any tree, each internal node v with leaf set S_v satisfies $\mathrm{cost}_v \geq |\partial S_v|$, the boundary of S_v in G. Writing $b(k)$ for the isoperimetric profile of G (the minimal boundary over all k-vertex subsets), we prove:

Theorem 1 (Sum-form profile bound). *For every contraction tree,* $\mathrm{tc} \geq \log_2 F(n)$ *where* F *obeys the recursion* $F(1) = 0$, $F(k) = 2^{\,b(k)} + \min_{1 \leq j \leq k/2} [F(j) + F(k-j)]$.

Theorem 2 (Dyadic-window bound). *Descending from the root always into the larger child visits, for every dyadic window* $W_j = (n/2^{j+1}, n/2^j]$, *at least one internal node whose leaf-set size lies in* W_j*; these nodes are distinct, hence* $\mathrm{tc} \geq \log_2 \sum_j 2^{\min_{k \in W_j} b(k)}$.

Both theorems are validated computationally (exhaustively at small n, and on 3000 random trees). Both fail to beat the max-form cap (Fig. 3): on sycamore_m20 they reach 47.2 and 47.0 against the carving value 53. The reason is structural:

Observation 1 (Profile bounds are capped at the bisection width). *Any lower bound computed from the isoperimetric profile* $b(\cdot)$ *alone is at most* $\max_k b(k) + \log_2 \log_2 n$, *and* $\max_k b(k)$ *is the bisection width—strictly below the balanced-cut (carving) value whenever the profile dips off-center. On sycamore_m20 it does: we exhibit a triply-verified 141-tensor subset with boundary exactly 40, against temporal-slab boundaries of 53 (Fig. 4).*

The $47 \to 53$ gap lives in jointly-nested-cut structure that per-size minima discard; the $53 \to 61.5$ gap is the log-sum residual. A complementary measurement (Fig. 5) shows why search cannot exploit the residual either: reducing the largest contraction by one unit systematically multiplies the number of contractions just below the peak, so total flops behave as if conserved under profile reshaping. This certified null is the backdrop against which the wins of the next two sections should be read: they do not contradict it—they change the input.

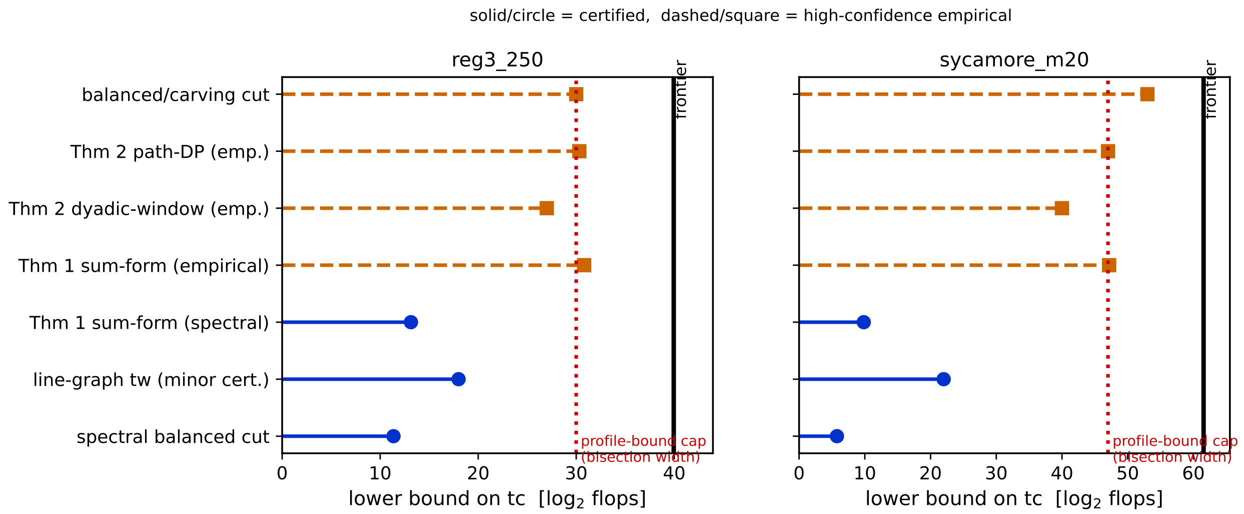

Figure 3: Lower-bound ladder for both closed instances. Solid stems with circles are certified (machine-checkable certificates); dashed stems with squares are high-confidence empirical values. Every profile-based bound (Theorems 1 and 2) is capped at the bisection width (dotted line, Observation 1); only the balanced/carving cut exceeds it. The solid vertical line is the optimizer frontier.

5 Beating the annealer I: structural simplification

The first mechanism that beats the tuned reference does not search better—it changes what is searched. Real-provenance networks carry large amounts of *rank-non-increasing* structure: tensors whose contraction with a neighbor can never increase any intermediate rank (rank-1 spiders, dangling legs, series chains). A preprocessing pass contracts this structure away greedily, the annealer runs on the reduced network, and the contracted fragments are spliced back into the final tree as fixed subtrees.

The effect is largest exactly where synthetic benchmarks never look. Figure 6a shows the shrink fraction across instances: 88.7% of the real Sycamore network's 3369 tensors are removed before annealing begins ($3369 \to 381$), 65.5% of surface-code d=21, 33.9% of nqueens_28, and 0% of the random 3-regular control—which is why the identical mechanism, correctly measured as useless on the pre-simplified synthetic proxy two cycles earlier, was nearly discarded. A matched-budget ablation on the real Sycamore network isolates the effect: the same anneal reaches tc = 61.3 with simplification and 76.0 without it, a $2^{14.7}$ ($\sim 27000\times$) difference from preprocessing alone. **The record march of Fig. 6b is our third main result**: three independently implemented simplify-then-anneal attempts drove the Sycamore record $60.605 \to 60.169 \to 60.104 \to 59.911$, and the mechanism transfers (surface-code and DBN records in Sec. 7 start from the same pass). The pass is exact—the contracted structure is rank-non-increasing, so no approximation is introduced—and costs milliseconds.

6 Beating the annealer II: waist surgery

The second winning mechanism attacks the annealer where it is provably blind. A converged annealed tree has a *waist*: the single dominant contraction that carries most of the log-sum (1). Local tree moves (rotations, subtree transfers) perturb the waist only through sequences of individually uphill steps, so late in the schedule the waist freezes. Waist surgery treats it globally instead: extract the bipartition induced by the dominant contraction, improve it as a graph-cut problem with bounded Fiduccia–Mattheyses passes over the tensor graph, rebuild the two sides cold with fixed-work sub-anneals, and splice. The move is accepted only if the rescored tree improves.

Instrumenting every surgery call yields **our fourth main result** (Fig. 7a): across 835 pooled calls on two instances, *the annealer's waist is never a locally minimal balanced cut.* On every single

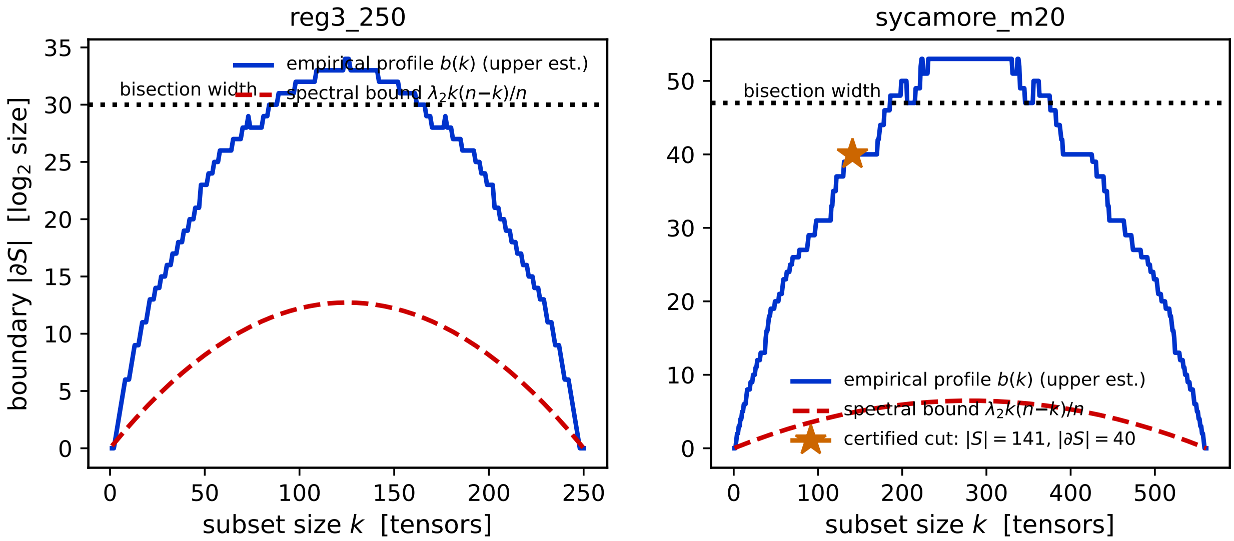

Figure 4: Isoperimetric profiles. The empirical profile $b(k)$ (solid) dips well below the bisection width (dotted) at off-center sizes; the certified star marks the 141-tensor sycamore_m20 subset with boundary exactly 40. This dip is what caps all profile-based bounds (Observation 1).

call, bounded FM found a strictly cheaper cut of comparable balance, by margins of 3–12 bits; the improved cuts are sparse rather than near-clique (clique fraction 0.01–0.05), i.e., genuine alternative separators, not degenerate rearrangements. The refinement is also load-bearing, not cosmetic: with surgery, the surface-code d=21 median improves from a ratchet-only 49.6 to 47.38, and the king-graph instance from 50.0 to 36.8. Fixed-work sub-anneals collapse run-to-run spread from 0.8 to 0.08 bits. Waist surgery set three simultaneous records (king graph 36.96, surface code 47.40, reg3_1000 131.07)—including the only record on an expander in the entire campaign—while leaving the two converged closed instances unchanged, exactly as the certified null of Sec. 4 predicts.

7 The record board, and what survives a change of machine

Figure 8 assembles the campaign's confirmed records against the tuned reference on all ten instances, colored by mechanism. The pattern, not the individual deltas, is the point: every record-setting mechanism injects global information as *input* to the local search (a simplified network, an improved cut, an elimination seed), and no record was ever set by replacing the annealer's control loop (Sec. 10). On four instances—including both certified closed instances and, notably, the 97-qubit random circuit and the 28-queens network—the tuned reference itself remains unbeaten.

Same-machine re-measurement. Records accumulated on one machine invite two objections: the gains might be machine-specific tuning artifacts, and single confirmed runs might not represent distributions. The dedicated-server campaign (Sec. 2) addresses both, and Fig. 9 shows its three headline panels. *Budget scaling* (Fig. 9a): on the real Sycamore network, waist surgery reaches 59.87 at 90 s while the tuned reference does not reach it even at 900 s (60.50)—a quality gap, not a speedup; on surface-code d=21, surgery at 90 s (47.39) matches the reference at 900 s (47.38)—a tenfold time-to-quality gain with both converging to the same floor. The picture is not universally rosy: on reg3_1000 the reference overtakes surgery at 900 s (129.8 versus 134.4), so the expander advantage is a fixed-budget effect. *Distributions* (Fig. 9b): over 15 repetitions, the surgery and reference distributions on surface-code d=21 are fully separated—surgery's worst run (47.91) beats the reference's best (48.18); king-graph and reg3_1000 medians improve by 3.0 and 2.4; the DBN seed mechanism shows a 14× tighter spread than the reference. One negative also reproduces: the Sycamore simplify-then-anneal pipeline of attempt 050, tuned against a faster development machine,

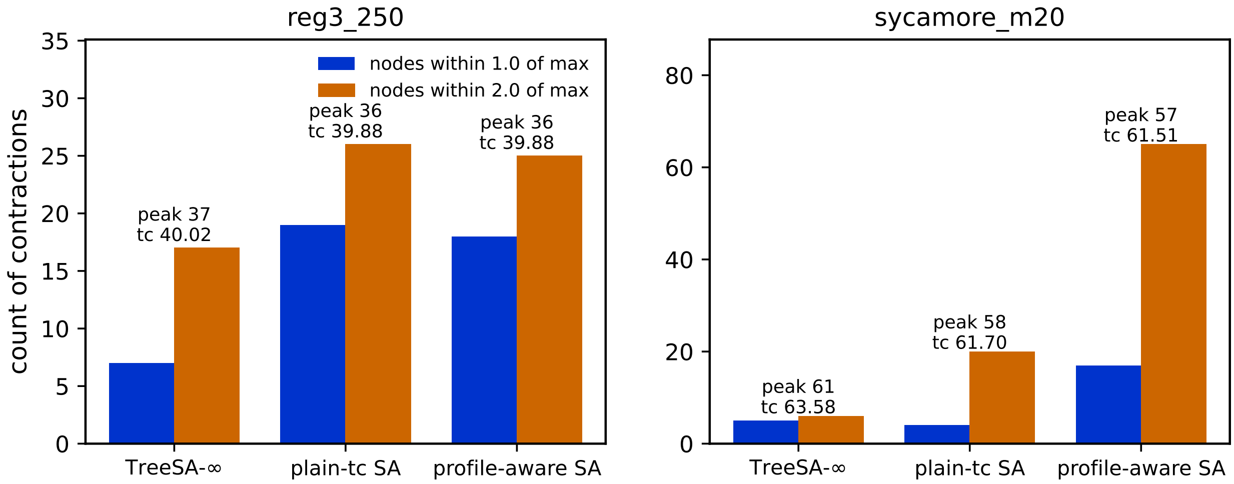

Figure 5: Cost-profile conservation at the frontier. Bars count contractions within 1.0 (solid) and 2.0 (hatched) of the largest; boxes give each tree's peak cost and tc. Trees that lower the peak by one unit acquire several times more near-peak contractions at essentially unchanged tc.

	this work (90 s)		Julia OMECO suite (same machine)	
instance	record	tuned ref.	best ≤90 s	best ≤900 s
sycamore_53_20_0	**59.91**	60.61	64.49	61.35
surfacecode_d21	**47.40**	48.76	48.98	48.75
ksg	**36.96**	39.12	39.22	38.27
reg3_1000	**131.07**	133.91	—	—
dbn_13	29.00	29.09	**28.03**	**28.03**
qft_27	29.61	29.61	29.62	29.62

Table 2: Records versus the reference Julia implementation (OMEinsumContractionOrders.jl) measured on the same server through its own harness. "Best $\leq B$" selects, over the configuration ladder (TreeSA at $\mathrm{sc}_{\mathrm{target}} \in \{20, \infty\}$ with 1/4/8 restarts, HyperND, treewidth back-ends), the best row whose measured runtime fits budget B. Bold marks the winner per instance. The dbn_13 row is won by the treewidth back-end (min-fill), instantly—see Sec. 8. reg3_1000 is not in the Julia suite. nqueens_28 is excluded: no ordering is stable at these budgets (see text).

is 0.5 *worse* than the reference at 90 s on the slower server and recovers only at 300 s—optimizer pipelines calibrate to absolute time, and honest cross-machine reporting must say so. *Family scaling* (Fig. 9c): on surface codes the surgery advantage grows with distance (+0.04, +0.02, +0.11, +1.04 at $d = 9, 13, 17, 21$) while its run-to-run spread stays collapsed; global-information injection pays off precisely at scale.

External baselines. Table 2 closes the loop against the reference Julia implementation, run on the same server through its own benchmark harness at a ladder of configurations, and against the suite's published numbers. Three observations. First, our records beat the best same-machine Julia configuration *at ten times our budget* on the three large structured instances (Sycamore 59.87 versus 61.35; surface code 47.40 versus 48.75; king graph 36.96 versus 38.27), and beat the entire published multi-optimizer suite by 6.8, 4.9, and 2.0 bits respectively. Second, the comparison is honest in both directions: on the DBN instance the published (and reproduced) treewidth row wins outright (28.03, instantly, versus our 29.00)—elimination methods are the right tool there, a theme Sec. 8 develops. Third, on nqueens_28 no stable ordering exists at this budget: our reference's fifteen-run spread on the server spans 104.9–159.3, single Julia runs land inside that spread, and we decline to claim a direction; distributional reporting is the only honest mode for that instance.

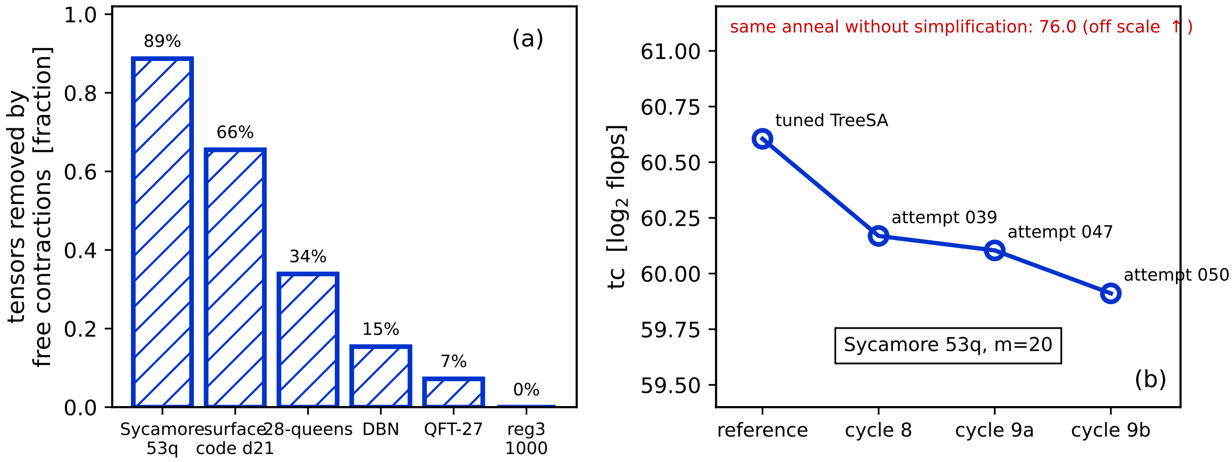

Figure 6: Structural simplification. (a) Fraction of tensors removed by the exact rank-non-increasing pass; the random-regular control is untouched at 0%. (b) The Sycamore 53q record march: the tuned reference (leftmost) and three successive simplify-then-anneal attempts. The same anneal without simplification reaches 76.0 (off scale): preprocessing, not search, carries the improvement.

8 Inference workloads: a frontier with two regimes

Contraction ordering is not a quantum-only problem: exact probabilistic inference contracts the tensor network of a graphical model, and TensorInference.jl [11] does so literally, optimizing an einsum whose inputs are one unity tensor per variable plus one tensor per factor, at full cardinalities. We exported all 65 instances of the UAI-2014 MAR benchmark from the package's own artifact in exactly that form, ranked them by a 5-second hardness scan, and measured the ten hardest (three dense deep-belief networks, four pedigree-linkage networks, one CSP, one grid, plus the package's own benchmark instance Promedus_14) under the full protocol: our four methods at five repetitions each, against a same-machine Julia ladder that includes the package's two defaults (the signature default GreedyMethod and the benchmark configuration TreeSA(ntrials=1, niters=5)), HyperND, the min-fill treewidth back-end, and tuned TreeSA.

Figure 10 shows every method's distance above the per-instance frontier, and **this figure is our fifth main result**. Three findings. *First, the shipped defaults are* 12–30 *bits off frontier* (4 000× to 10^9× in flops): GreedyMethod lands 14.6–16.6 above on the DBN family and 30.5 above on linkage_15; the benchmark TreeSA configuration is little better there (10–16 above). Whatever mechanism one prefers, an inference user who accepts defaults pays orders of magnitude for it. *Second, the frontier itself splits into two structural regimes.* On dense DBNs, elimination methods win outright: min-fill treewidth and HyperND reach the frontier (25.9–28.0) in milliseconds, tuned annealing trails by 5–8 bits, and our best method (the elimination-seeded annealer, Sec. 9) lands within 0.7 of the elimination frontier with a 10× tighter spread than the plain reference. On pedigree linkage the regimes invert: min-fill is 6–15 bits *above* the frontier, HyperND's back-end (KaHyPar) segfaults on all four instances, and annealing-family methods—ours leading on all four—hold the frontier. *Third, the regime boundary is a structure property, not a size property*: the DBN networks have 44–66 shared labels of high degree (elimination's sweet spot), while linkage networks are sparse with a thousand variables. A portfolio that probes elimination first (milliseconds) and falls back to annealing otherwise captures the frontier everywhere we measured; our simplification pass (Sec. 5) additionally absorbs the per-variable unity tensors that inference-style einsums carry by construction.

9 Composites, the anytime axis, and elimination seeds

Three secondary mechanisms round out the toolbox. *Composites*: the winning mechanisms are complementary (simplification transforms the input, surgery escapes waist freezing, both feed the

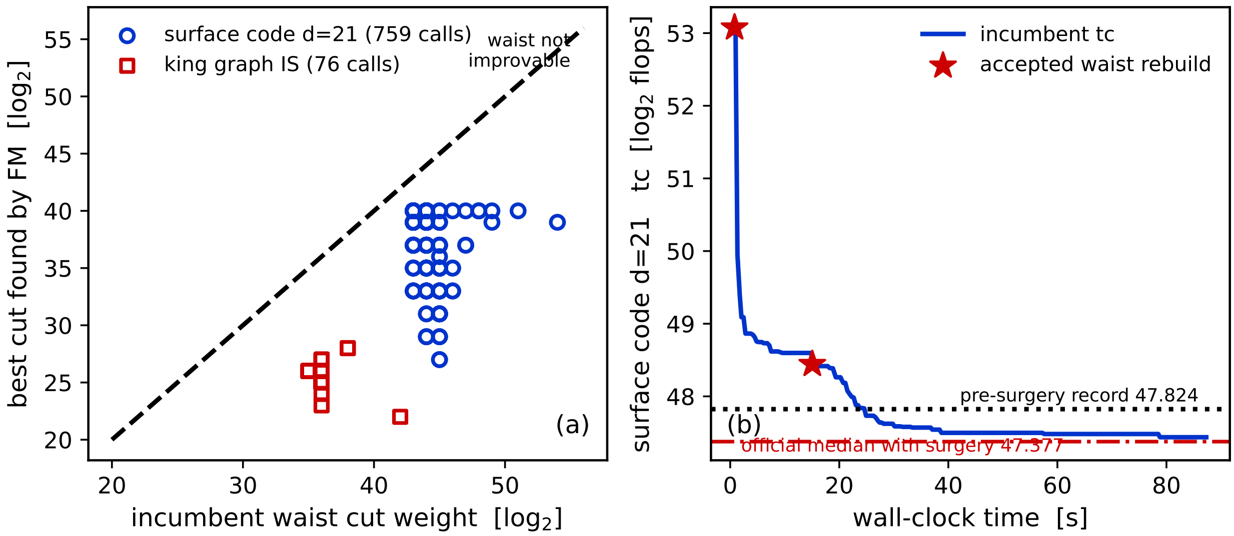

Figure 7: Waist surgery. (a) Every instrumented surgery call: incumbent waist cut weight versus the best comparable-balance cut found by bounded FM. All 835 points lie strictly below the diagonal—the annealed waist is never locally minimal. (b) A representative surface-code d=21 run: incumbent tc (solid), accepted waist rebuilds (stars), the pre-surgery record (dotted) and the official with-surgery median (dash-dotted).

same annealer), and a composite that routes between them set three records in a single run, including the first fall of an expander instance. Racing arms with budget-fraction schedules matter here; a probe-leader policy never won. *The anytime axis*: with the harness clocking candidate output over time, records exist for time-to-frontier as well: 16.1 s to reach the reg3_250 frontier (versus 20.2 s for the reference), and 0.2 s on the DBN instance. *Elimination seeds*: on hypergraph-like instances with few, high-degree labels, seeding the annealer with a min-fill variable-elimination order gives it a start (tc = 28.8 at 0.2 s on dbn_13) that plain greedy construction (44.7) cannot approach; the mechanism's boundary is equally sharp—at 4086 labels (nqueens) the VE peel freezes and the seed is useless. This is the method that holds the DBN-regime rows in Sec. 8.

10 What does not work: a mechanism taxonomy

Twenty further mechanisms were implemented, measured, and falsified under the same protocol. **Their taxonomy is our sixth main result**, because the failures have measured causes that organize into four families (Table 3), and the pattern they draw is the paper's central lesson stated negatively: *every mechanism that injected global information as search control failed; every mechanism that injected it as input won.* Construction-time global structure (separators, spectral cuts, boundary combs) loses to the greedy portfolio it seeds; search-control globality (targeting, tempering, blocking, acceptance relaxation) breaks the annealer's schedule invariants; exhaustive enumeration hits certified combinatorial walls (window-exact optimality of the incumbent frontier; block counts exploding past 10^5 within bits of the minimum boundary—with an outer-product counterexample showing the connectivity-restricted variant is not even correct); and population policies drain a small recombination reservoir (about ten productive recombinations) regardless of diversity management.

11 Methodology: a research loop that distrusts itself

The results above were produced by an autonomous research loop—56 optimizer attempts over ten cycles, each in an isolated worktree with a pre-registered hypothesis, scored by a sandboxed validator—and the loop's central design lesson is that its most important outputs were *retractions*

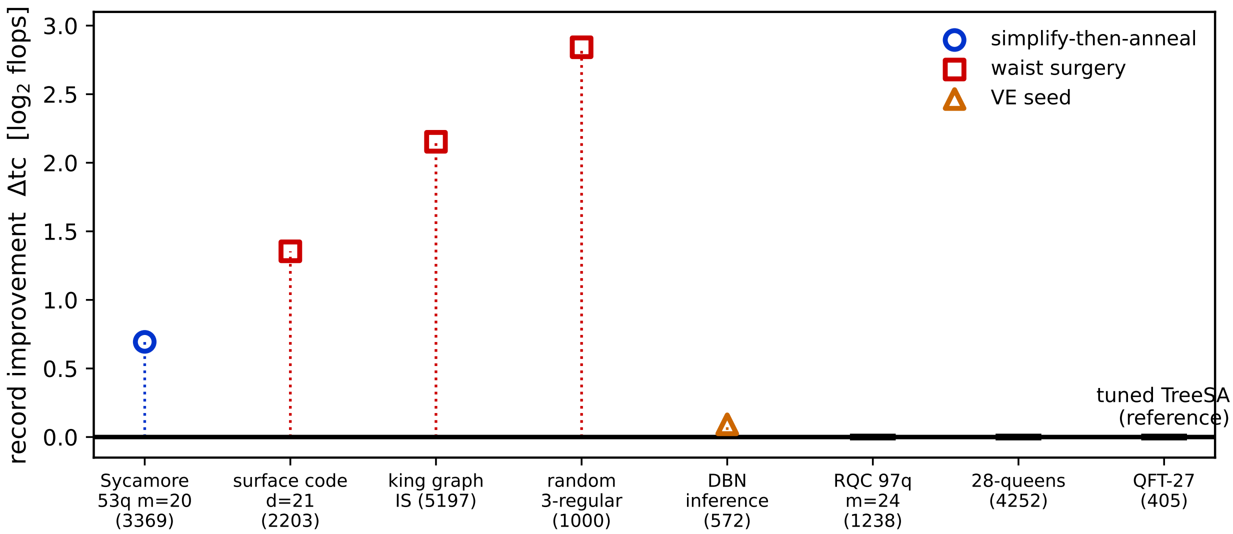

Figure 8: The record board: confirmed improvement over the tuned TreeSA reference on each instance (validator protocol: confirm-twice, worse-of-two, per-run relabeling). Marker shape and color encode the winning mechanism; horizontal dashes mark instances where the tuned reference is the record holder.

(Fig. 11). Three failure modes were caught by the harness rather than by luck:

(i) Schedule exploitation. The first gate (a speed-versus-baseline bar) was passed by budget hygiene alone—anytime writes and early termination—with no algorithmic content. The fix was structural: fold the exploit into the reference, making schedule-only candidates score zero by construction, and keep the exploit as a permanent negative control.

(ii) Mis-tuned-baseline gains. Eight mechanisms then "beat" the references by up to 800×—until a strengthened reference (the one-line $\mathrm{sc}_{\mathrm{target}}$ fix of Sec. 3) reclaimed the entire gain. What survived was the tuning discovery itself, and the leaderboard practice of ratcheting references to absorb every discovered improvement.

(iii) Instrument bias. A failed attempt's diagnosis revealed that the independent scorer silently rejected contraction trees deeper than ∼475 (a recursion limit), biasing the admissible space against exactly the deep, MPS-like trees that suit circuit instances. The scorer was fixed, its negative-control suite re-passed, and the affected hypothesis re-tested.

As the campaign scaled, the harness grew with it: parallel scoring lanes with per-child CPU accounting, median-of-three scoring above 1000 tensors, early abort of hopeless confirmation runs, and harness-clocked anytime curves that make time-to-frontier claims backdate-proof. The pattern that generalizes: score only through a hardened independent validator; maintain executable negative controls, and when a new exploit is found, patch the harness and add the exploit to the control suite; ratchet references so yesterday's wins are tomorrow's baselines; and report distributions wherever single runs are noisy. Under this regime, every claim in this paper is either machine-certified, or labeled high-confidence with its evidence, or was retracted before publication—and the audit trail (per-attempt logs, certificates, and cycle reports) ships with the code.

12 How the results were found: steered autonomous research

The findings of this paper were produced by an autonomous research loop (machine-proposed hypotheses, isolated implementations, hardened scoring) that was *steered* by a small number of human decisions. Because the loop retained every artifact, the causal path from the opening question to each main result can be reconstructed exactly. Figure 12 shows that path left to right: the top row quotes the steering prompts, the middle row shows only the attempts that carried results forward (most of the 56 attempts served as controls, falsifications, or dead branches feeding Secs. 10 and 11),

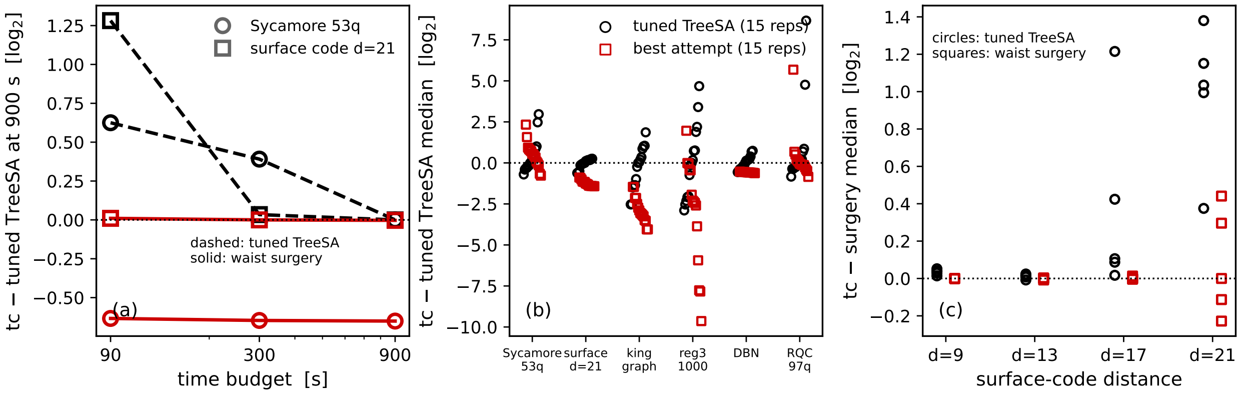

Figure 9: Same-machine matched-budget campaign (dedicated 2-core server; all runs single-threaded). (a) Median tc versus budget, anchored per instance at the tuned reference's 900 s median: on Sycamore the surgery gap never closes; on surface code d=21 surgery at 90 s already sits on the reference's 900 s floor. (b) All 15 repetitions per instance at 90 s, centered on the reference median: circles are the tuned reference, squares the best attempt per instance. (c) Surface-code family: repetitions at 90 s versus code distance, anchored at the surgery median per distance.

and each arrow states why the transition happened.

Three properties of this trajectory are worth stating explicitly. *First, every pivot was forced by a measurement, not chosen freely.* The move from tc-optimization to the anytime axis happened because thirteen mechanisms and a second toolchain had converged on one frontier that was then certified locally optimal three independent ways; the move to scale happened because time-to-frontier proved heavy-tailed at the claim threshold; the move to global-information mechanisms happened because every search-control variant had failed with a structural cause while every input-level variant had won. *Second, the human contributions were few but load-bearing*: the demand for a strengthened reference (which produced the tuning result of Sec. 3), the switch to real-provenance instances (which unlocked the simplification pillar—the same mechanism had been correctly measured as useless on the pre-simplified proxy two cycles earlier), the question "can global information help the local search escape?" (which became waist surgery), and the directive to demonstrate the methods on inference workloads (which produced the two-regime result of Sec. 8). *Third, retests are cheap when provenance changes*: attempt 013's null on the synthetic proxy and attempt 039's records on the real circuit are the same mechanism measured on different inputs, and only the pre-registered instance provenance made the contradiction interpretable rather than embarrassing.

13 Conclusions

When can tuned simulated annealing be beaten at contraction ordering? The campaign's answer is a characterization. *Never, before tuning*: one shipped memory-target default costs 20× to 2100×, more than every mechanism difference we measured, and comparisons against untuned incumbents are not informative. *Never, on small converged instances*: thirteen paradigms and two toolchains tie a frontier that is certifiably width-optimal, with the optimum localized to $[53, 61.5]$ on the circuit proxy and a proof that profile-based bounds can go no further. *Yes, on large structured instances*—but only, in our campaign, by mechanisms that inject global structure as input to the local search: exact simplification (up to 89% of tensors removed, $2^{14.7}$ from preprocessing alone on Sycamore) and waist surgery (the annealed waist is never a locally minimal cut; three records, distribution-separated on surface code, with an advantage that grows with code distance). *And sometimes the answer is to stop annealing*: on dense inference networks elimination orderings win outright, annealing wins the sparse pedigree regime, and the shipped defaults of an inference package sit 12–30 bits above either—so a milliseconds-cheap elimination probe belongs in front of every annealer.

Deliverables: the tuned defaults, the exact simplification pass, the waist-surgery refiner, a warm-

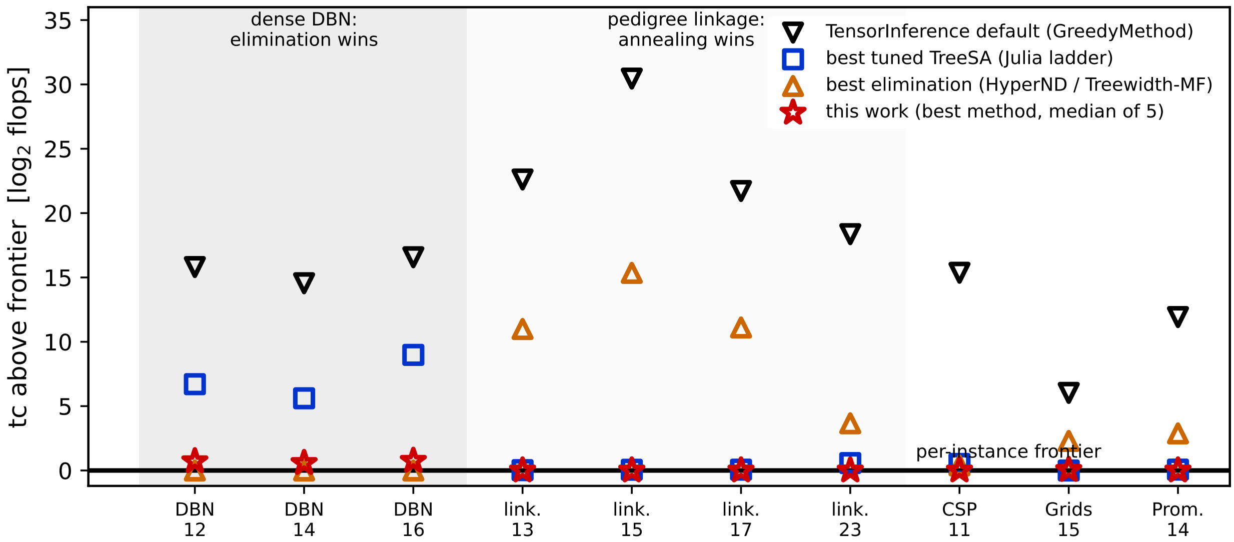

Figure 10: The ten hardest UAI-2014 MAR instances (from a 65-instance scan): each method's tc above the per-instance frontier, same machine, 90 s budgets for search methods. Down-triangles: TensorInference.jl's signature default. Squares: best tuned TreeSA from the Julia ladder. Up-triangles: best elimination method (min-fill treewidth / HyperND; KaHyPar segfaults on all four linkage instances, so its rows there are min-fill only). Stars: this work (best method, median of five). Shaded bands mark the two regimes.

start annealing API, and two bug fixes are released in the omeco library (with the simplification and surgery passes prepared for upstreaming to OMEinsumContractionOrders.jl), together with the validator, all certificates, and the 56-attempt audit trail. Open problems: carving-width-type lower bounds (profile methods are provably exhausted); the two instances that repelled every mechanism (the 97-qubit random circuit and 28-queens, the latter unstable at practical budgets); heterogeneous bond dimensions; and sliced contraction, where the interaction between slicing and both winning mechanisms is unexplored.

Code and data availability

The optimizer library (omeco), the validation harness, all certificates (treewidth minors, cut sets, and their verification scripts), per-attempt logs, cycle reports, the same-machine campaign data (305 + 200 + 80 + 30 scored runs and the Julia ladder), and the UAI instance exporter are available in the project repository.

Use of AI systems

This research was conducted as a steered autonomous research campaign: LLM-based agents proposed hypotheses, implemented attempts in isolated worktrees, and drafted analyses, under human steering decisions documented verbatim in Sec. 12 and under a validation protocol in which no agent-reported number is trusted—every scored quantity is recomputed by an independent checker from artifacts, and all certified claims are verified by machine-checkable certificates independent of any agent output. The manuscript was drafted with the same agents and revised by the human author, who takes full responsibility for its content.

family / mechanism (attempt)	measured cause of failure
Construction proxies	
BFS/planar separators (037)	seeds lose to greedy portfolio even with separator injection
spectral bisection (041)	power iteration cannot converge (gap $\sim 1/n^2$); a perfect bisection is still dominated
boundary-MPS combs (036/044)	cutwidth 90–126 lower-bounds any comb, far above balanced-tree records
mass randomized greedy (040)	quality ceiling 15–208 bits above frontier at 1.2 builds/s
Search-control globality	
elimination-order annealing (045)	flat landscape: 30 improvements per 2.5M moves; representation floor $\sim 2\times$ above tree space
congestion-softmax targeting (042)	collapses onto the dominant node on peaky profiles; 40–80% dead traversal
cache-blocked regional SA (043)	pointer trees have no memory contiguity; blocking starves global moves
replica-exchange tempering (048)	swap acceptance $\to$ 0% at scale (energy gaps grow with n)
late-acceptance hill climbing (051)	never cools on plateau-dominated landscapes; schedules are load-bearing
MCTS over prefixes (050)	reaches depth 3 of $\sim$380 at 8–25 rollouts/s
Exhaustive enumeration	
window-exact splice (033)	frontier trees are window-exact optimal ($\leq$ 16 leaves); the dominant window spans a constant fraction
Tamaki-style block DP (056)	feasible low-boundary blocks explode past 10^5–10^6 within bits of $\min_W$; outer-product counterexample falsifies the connectivity-restricted variant
Population policy	
path-relinking / consensus / diversity pools (046/049/055)	basin complementarity drains after $\sim$10 recombinations regardless of pool policy

Table 3: The falsified-mechanism taxonomy. Each row is an implemented, scored, and diagnosed attempt; attempt numbers index the public audit trail. The four families fail for four distinct measured reasons, and none of the failures is a tuning accident.

Acknowledgments

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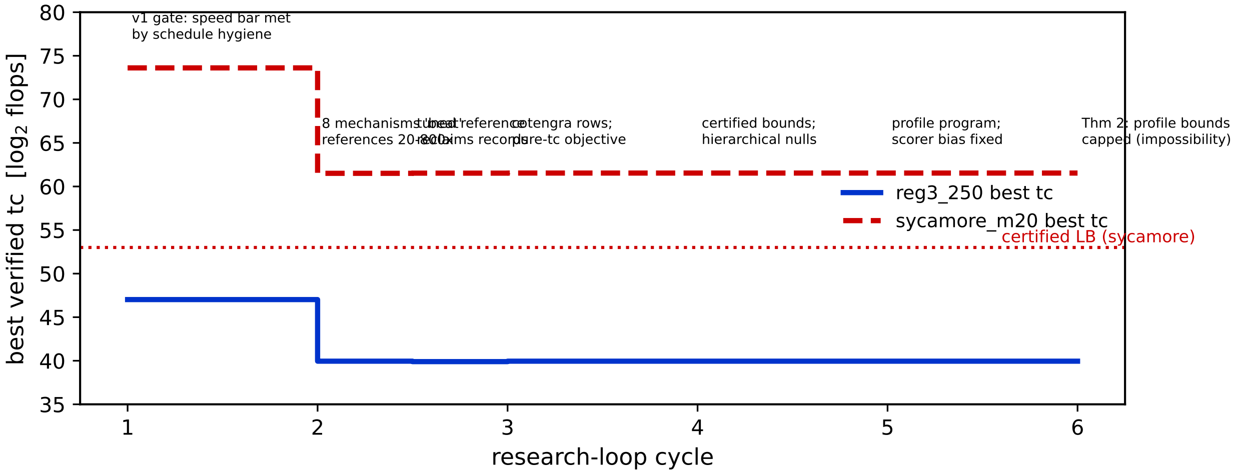

Figure 11: Best verified tc over the research-loop cycles. Apparent progress in cycle 2 is reclaimed by the strengthened reference (cycle 2.5); thereafter the closed-instance frontier is flat while the certified picture grows, and cycles 7–10 move the record board on the large structured instances instead. Annotations mark the three instrument-caught failure modes.

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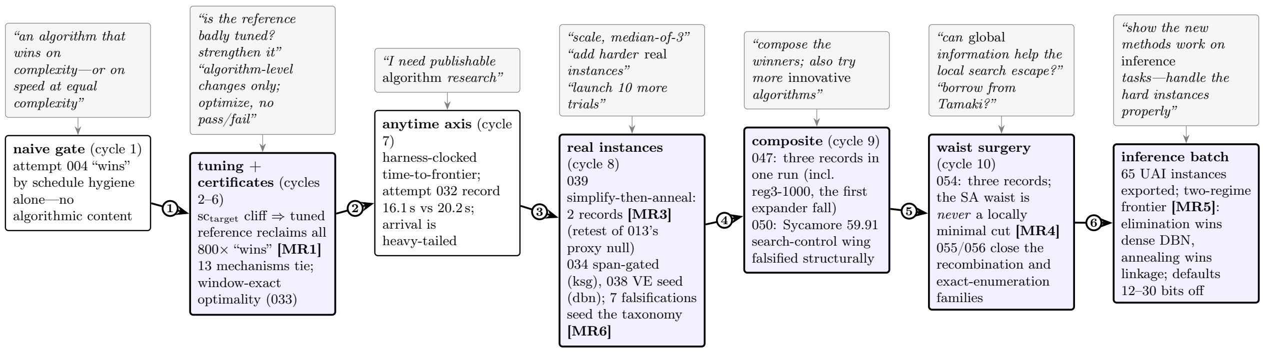

Figure 12: The research roadmap, left to right. Top row (grey): the human steering prompts, essentially verbatim. Bottom row: the result-carrying attempts only (shaded boxes contain main results); most of the 56 attempts are controls and falsifications that feed Table 3 and Sec. 11. Each numbered arrow is a transition forced by a measurement: (1) the gate was passed without an algorithm, so the gate was hardened and references ratcheted; (2) tc saturated and was certified locally optimal, so the objective axis changed to arrival time; (3) speed gains sat at the confirmation noise floor while scale budgets were non-converged, so the regime changed; (4) the winning mechanisms proved complementary, so they were composed while exploration continued; (5) every win injected global information as *input* while every search-control variant failed, so global information was aimed directly at the local search's blind spot—the dominant cut; (6) the mechanism portfolio was complete, so it was pointed at a second application domain to test transfer.